AFF–1014

Sub. Code
4MPH1C1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Physics

MATHEMATICAL PHYSICS — I

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. When is a vector said to be

   (a) Solenoidal

   (b) Irroational?

2. For a position vector $\vec{r} = \hat{i}x + \hat{j}y + \hat{k}z$, show that $\nabla \times \left(\frac{\hat{k}}{r}\right) = \frac{-\hat{i}y + \hat{j}x}{r^3}$.

3. Define inner products of a vector.

4. Prove that A and A^T have the same eigen values.

5. Transform the equation $xy'' + y' + 1 = 0$ into a linear equation with constant coefficients and hence solve it.

6. How will you identify a second order linear differential equation in y with variable coefficients that has

   (a) $y = e^x$ as a solution

   (b) $y = e^{-x}$ as a solution?

7. If $u + iv$ is analytic, show that $v + iu$ is not analytic.

8. Define removable singularity with an example.

9. State and explain Dirichlet's theorem.

10. Illustrate with example the physical significance of Fourier series.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) If $\vec{F}$ is a vector point function, then $\text{curl}\left(\text{curl}\,\vec{F}\right) = \nabla \times \left(\nabla \times \vec{F}\right) = \nabla\left(\nabla.\vec{F}\right) - \nabla^2\vec{F}$.

Or

(b) If $\vec{F} = xy\vec{i} - z\vec{j} + x^2\vec{k}$, evaluate $\int_C \vec{F} \times d\vec{r}$, where C is the curve $x = t^2$, $y = 2t$, $z = t^3$ from $(0,0,0)$ to $(1,2,1)$.

12. (a) Show that the vectors $X_1 = (1, -1, -2, -4)$, $X_2 = (2, 3, -1, -1)$, $X_3 = (3, 1, 3, -2)$ and $X_4 = (6, 3, 6, -7)$ are linearly dependent. Find also the relationship among them.

Or

(b) Show that the eigen values of a Hermitian matrix are all real.

13. (a) Solve the equation $p(p + y) = x(x + y)$.

Or

(b) Solve the equation $x\frac{d^2y}{dx^2} + (2 - x)\frac{dy}{dx} - y = 0$, given that $y = \frac{1}{x}$ is a solution.

14. (a) Show that the function $f(z) = \sqrt{|xy|}$ is not regular at the origin, although Cauchy-Riemann equations are satisfied at the origin.

Or

(b) State and prove Cauchy's integral theorem.

15. (a) Find the Fourier series of the function e^x in the interval $-\pi < x < \pi$.

Or

(b) State and prove Parseval's relation for Fourier transformation.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. (a) State and prove Stoke's theorem.

(b) Prove that $\int_C \vec{r} \times d\vec{r} = 2\iint_S d\vec{S}$

Where S is a diaphragm enclosing a circuit C.

17. State Cayley-Hamilton theorem for matrices. Use it to find the inverse of the matrix

$$\begin{bmatrix} 1 & -1 & 0 \\ -1 & 0 & 2 \\ -2 & 1 & 1 \end{bmatrix}.$$

18. Using Frobenious method, solve Sturm-Liouville differential equation.

19. Apply calculus of residues to show that

$$\int_0^{2\pi} \frac{d\theta}{a + b\cos\theta} = \frac{2\pi}{\sqrt{a^2 - b^2}}, \; a > b > 0.$$

20. Define a Fourier integral and prove that Fourier integral theorem.

$$f(x) = \frac{1}{\pi} \int_0^{\infty} du \int_{-\infty}^{\infty} \cos u(x - t) f(t)\, dt$$

where $f(x)$ is a real function.

AFF–1015

Sub. Code
4MPH1C2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Physics

CLASSICAL DYNAMICS OF RELATIVITY

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What are generalised co-ordinates? How are they different from our usual Cartesian co-ordinates?

2. State the principle of virtual work.

3. Show that central force motion takes place always in a plane.

4. What are Kepler's laws of Planetary motion?

5. Show that the conjugate momentum corresponding to a cyclic coordinate is conserved.

6. How many conventions are possible in the definition of Euler angles?

7. What is the use of generating function is canonical transformation?

8. Find the value of $\lfloor P_z, L_y \rfloor$.

9. When do you say than an equation is in covariant form?

10. Write the postulates of special theory of relativity.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) State D'Alembert's principles and hence the Lagrange's equations of motion for a system of particles free from constraints.

Or

(b) Obtain equation of motion for a simple pendulum by Lagrange's method.

12. (a) State and prove Bertrand's theorem.

Or

(b) Derive the differential equation for the orbit of a particle moving under central force.

13. (a) Show that the kinetic energy of a rigid body can be expressed as $T = \frac{1}{2}$ IW.

Or

(b) What are dispersion and wave packet? Explain briefly.

14. (a) Discuss the harmonic oscillator problems as an example of Hamilton-Jacobi method.

Or

(b) Obtain the Lagrange's equation of motion for small oscillations of a system in the neighbourhood of stable equilibrium.

15. (a) Explain Minkowski's four dimensional space.

Or

(b) Establish the relation giving variation of mass with velocity of a particle.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Prove the laws of conservation of linear momentum, angular momentum and energy for a system of interacting particles.

17. What is differential scattering cross-section? Discuss the problem of scattering of charged particles by a Coulomb field and obtain Rutherford's formula for the differential scattering cross-section.

18. State and prove the principle of least action.

19. Obtain the equation of motion for a heavy symmetric top with one point fixed.

20. Write down the Maxwell's equations and show that they are invariant under Lorentz transformation.

AFF–1016

Sub. Code
4MPH1C3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Physics

ADVANCED ELECTRONICS

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is the distinction between an intrinsic and extrinsic semiconductors?

2. List the differences between depletion mode and enhancement mode.

3. Draw the circuit diagram of an op-amp differentiator.

4. Define input offset voltage and explain why it exists in all op-amps.

5. A shift register has eight flip-flops. What is the largest binary number that can be stored in it?

6. Distinguish between RAM and ROM.

7. What is meant by resolution of an A/D conrerter?

8. List any four applications of a Schmitt trigger.

9. What is meant by single – side band suppressed carrier modulation?

10. Define Modulation index.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Derive an expression for the carrier concentration of an intrinsic semiconductor.

Or

(b) Define FET parameters μ, g_m and r_d. Obtain relations between them.

12. (a) Draw the non-inverting amplifier circuit of an op-amp in closed – loop configuration and obtain the expression for the closed – loop gain.

Or

(b) Explain how band – pass filter can be constructed using op-amp.

13. (a) Explain with logic diagrams and truth table the working principle of a JK flip-flop.

Or

(b) Write short note on Charged Coupled Devices (CCD).

14. (a) Explain the working of op-amp as a Schmitt trigger using input and output wave forms with circuit diagram.

Or

(b) Explain the working of binary weighted resistor type D/A converter.

15. (a) Explain the Generation and Demodulation of Pulse Position Modulation (PPM).

Or

(b) Discuss the principle and working on Magnetrun.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Explain the construction, biasing, operation and V-I characteristics of SCR.

17. Explain the working of op-amp as a comparator using input and output wave forms with circuit diagram.

18. What are the different kinds of logic families available? Explain how a DTL NAND gate works, with a suitable diagram.

19. Discuss the applications of astable multivibrator as a

(a) Square Wave oscillator

(b) Free running ramp generator.

20. Describe the frequency spectrum for sinusoidal AM. Calculate the average power, effective voltage, and current for sinusoidal AM.

AFF–1017

Sub. Code
4MPH1C4

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Physics

SOLID STATE PHYSICS

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Clarify the following unit cell into proper crystal system $a = 0.95\ nm$, $b = 1.1\ nm$, $c = 0.5\ nm$ with $\alpha = 36^\circ$, $\beta = 80^\circ$ and $\gamma = 95^\circ$.

2. If 0.28 nm is the spacing between the nearest neighbouring ion in NaCl lattice, what is the unit cell parameter?

3. What is the cut-off frequency for a linear mono atomic lattice? Velocity of sound = 3×10^3 m/sec and atomic spacing $(a) = 3 \times 10^{-10}$ meter.

4. To what type of force, the two branches respond in the case of diatomic lattice?

5. What is measured by the periodicity in de Hass-Van Alphen effect?

6. In first Brillouin zone, how the effective mass varies with its position on the E–K curve?

7. How the shape of hysteresis loop of a ferroelectric material changes as the temperature is increased?

8. What is the cause of dielectric relaxation?

9. Why is the electric field zero inside a super conductor?

10. What is a Bloch Wall?

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) What is Bravais lattice? What are the different space lattices in the cubic system? How many lattice points per unit cell are there in each of these lattices?

Or

(b) Show that the reciprocal lattice to bcc is fcc.

12. (a) Establish relation between elastic compliances and stiffness constants of a cubic crystal.

Or

(b) Discuss the inelastic scattering of phonons.

13. (a) Bring out the origin of the Energy gap and also find its magnitude.

Or

(b) Explain the number of orbitals in a band.

14. (a) Explain electronic polarization in atoms and obtain an expression for electronic polarizability in terms of radius of atom.

Or

(b) Give an account of the quantum theory of paramagnetism.

15. (a) Give an account of BCS theory of super conductivity.

Or

(b) Discuss in detail Ferromagnetic domain.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Derive the structure factors of BCC and FCC lattices.

17. Discuss the vibrations of mono atomic lattices and derive the dispersion relation. Also give the plot of W versus K and explain it.

18. Explain about Hall effect and obtain an expression for Hall coefficient.

19. Describe the paramagnetic susceptibility of conduction electrons and hence derive the Pauli spin magnetization.

20. Explain the thermodynamics of super conducting transitions and arrive London equations.

AFF–1018

Sub. Code
4MPH1E1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Physics

***Elective* — NUMERICAL METHODS**

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define experimental error, truncation error and rounding off error.

2. Write down the normal equations to fit a straight line and parabola.

3. Write down Newton-Raphson's formula for iteration.

4. Explain 'pivoting' and what are its two types.

5. Define 'Divided differences' and write down the formula.

6. Write down Newton's forward difference and backward difference interpolation formula.

7. Write down the trapezoidal's rule to evaluate $\int_{x_0}^{x_n} f(x)dx$.

8. State the basic principle for deriving Simpson's 1/3rd rule.

9. Write down the Euler's algorithm to the differential equation $\frac{dy}{dx} = f(x, y)$.

10. Write down the Rune-Kutta formula of fourth order to solve $\frac{dy}{dx} = f(x, y)$ with $y(x_0) = y_0$.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Fit a straight line $y = ax + b$ to the data given below by the method of group averages.

x :	0	5	10	15	20	25
y :	12	15	17	22	24	30

Or

(b) Fit a straight line by least squares method for the data given below. Also estimate the value of y at $x = 2.5$.

x :	0	1	2	3	4
y :	1	1.8	3.5	4.5	6.3

12. (a) Find a positive root of $xe^x = 2$ by the method of False position.

Or

(b) Find an iterative formula to find the reciprocal of a given number N and hence find the value of $\frac{1}{19}$. Write a C program for it.

13. (a) A function $f(x)$ is given by the following table. Find $f(0.2)$ using Newton's forward interpolation formula.

x :	0	1	2	3	4	5	6
$y = f(x)$:	176	185	194	203	212	220	229

Or

(b) Using Lagrange's interpolation formula find the equation of the cubic curve.

x :	–1	0	2	3
$y = f(x)$:	–8	3	1	2

14. (a) The velocity 'v' of a particle at distances from a point on its path is given below. Estimate the time taken to travel 60 m using :

(i) Simpson's 1/3rd rule

(ii) 3/8th rule.

Distance s (in m) :	0	10	20	30	40	50	60
Velocity V (in m/sec) :	47	58	64	65	61	52	38

Or

(b) The population of a town in shown below. Find its rate of graph for the year (96) using Numerical differentiation.

Year :	1931	1941	1951	1961	1971
Population (in thousand) :	40.6	60.8	79.9	103.6	132.7

15. (a) Solve $\frac{dy}{dx} = y + e^x$, given $y(0) = 0$ for $x = 0.2, 0.4$ using improved Euler's method.

Or

(b) Find the values of $y(1.1), y(1.2)$ using Runge-Kutta's third order method for the differential equation $\frac{dy}{dx} = y^2 + xy,\ y(1) = 1$.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Fit a second degree parabola to the following data using the method of least squares

x :	1.0	1.5	2.0	2.5	3.0	3.5	4.0
y :	1.1	1.3	1.6	2.0	2.7	3.4	4.1

17. Solve the following system of equations using Gauss-Jordan's method :

$4x - y - z = -7;\ x - 5y + z = -10;\ x + 2y + 6z = 9$.

18. (a) Apply Gauss's forward central difference formula and estimate $f(32)$ from the table given below : (6)

x :	25	30	35	40
$y = f(x)$:	0.2707	0.3027	0.3386	0.3794

(b) Write a 'C' program for this problem. (4)

19. Find the first and second derivatives of $\sqrt{x}$ at $x = 15$ for the table given below :

x :	15	17	19	21	23	25
$\sqrt{x}$:	3.873	4.123	4.354	4.583	4.796	5.0

20. Using Runge-Kutta's fourth order method solve $\frac{dy}{dx} = \frac{y^2 - x^2}{y^2 + x^2}$ with $y(0) = 1$ at $x = 0.2, 0.4$.

AFF–1019

Sub. Code
4MPH1E2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Physics

Elective — CRYSTAL GROWTH AND THIN FILMS

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. List out the methods of low temperature solution growth.
2. What criteria should be adopted in the selection of crystallizer?
3. Define physical gel. Give examples.
4. What is complex-decomplexion method? Give example.
5. Name the factors to be considered for the growth of crystals from the melt.
6. Define chemical vapour transport reaction.
7. List out the various techniques involved in the preparation of thin films.
8. Name the various sources in the distribution of deposit.
9. Name the two different ways of registering the diffracted X-rays by powder method.
10. State and explain the Laue's equations.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Describe a constant temperature bath of growing a crystal with a diagram.

Or

(b) Write a note on seed preparation, mounting and seasoning.

12. (a) Describe double diffraction method gel growth.

Or

(b) Explain the principle of gel growth.

13. (a) Explain about physical vapour deposition with a block diagram.

Or

(b) List out the various ways of promoting chemical vapour transport reaction method.

14. (a) Write a note on R.F. sputtering method.

Or

(b) Explain about vapour plating method of film deposition.

15. (a) Explain the theory of Bragg's law and obtain the Bragg's equation.

Or

(b) Calculate the glancing angle of the (1, 1, 0) plane of a simple cubic crystal with a = 2.814 Å corresponding to the second-order diffraction maximum for X-rays of wavelength 0.710 Å.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Discuss the following :

    (a) Measurement of super saturation (3)

    (b) Meir's solubility diagram (4)

    (c) Slow cooling method. (3)

17. (a) Explain about chemical reduction method (4)

    (b) List out the various advantages of gel method. (6)

18. (a) Describe briefly the liquid encapsulation in crystal growth process. (4)

    (b) Describe in depth the chemical vapour deposition method in crystal growth. What is the uniqueness of this method over the physical vapour deposition method? (6)

19. (a) Explain electro Deposition method. (6)

    (b) Write a note on deposition by Chemical Reaction. (4)

20. (a) Explain the spectroscopic techniques used in UV spectroscopy with emphasis on sources, detection and investigations. (6)

    (b) What are the applications of U-V radiation? (4)

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AFF–1020

Sub. Code
4MPH2C1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Physics

MATHEMATICAL PHYSICS – II

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. State the conditions for the existence of Laplace transform of a function.

2. State linearity theorem for Fourier transform.

3. Write Laplace equation in cylindrical and spherical polar coordinates.

4. Give the physical interpretation of D'Alembert's solution of vibration string.

5. If velocity and speed are tensors then what are their ranks?

6. What is Einstein's summation convention?

7. Define permutation groups.

8. What are the various symmetry operations involved in C_{2V} point group?

9. State the orthogonal properties of Legendre polynomials. Hence find the value of $\int_{-1}^{+1} [P_4(x)]^2 dx$.

10. Find the probability that a card drawn at random from an ordinary pack is one of the court – cards.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) State and prove convolution theorem in Fourier transform.

Or

(b) Find the inverse Laplace transform of $\frac{1}{s^2 + a^2}$.

12. (a) Find the solution of Laplace's equation in Cartesian coordinates using the method of separation of variables.

Or

(b) A string of Length l fixed at both ends is plucked at a distance d from one fixed point by an amount h. Find the displacement at any point on the string at any instant of time.

13. (a) Obtain the tensor form of gradient and divergence.

Or

(b) If A^{μ} and B_{υ} are the components of a contravariant and covariant tensors of rank one, show that $C^{\mu}_{\upsilon} = A^{\mu} B_{\upsilon}$ are the components of a mixed tensor of rank two.

14. (a) Mention any six properties of classes.

Or

(b) What is an abelian group? Mention its properties is a cyclic group abelian.

15. (a) If two normal universes have the same total frequency but the standard deviation of one is K times that of the other, show that the maximum frequency of the first is $\frac{1}{K}$ that of the other.

Or

(b) Prove that $H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n}(e^{-x^2})$.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. (a) Find the Fourier transform of the Gaussian distribution function $f(x) = Ne^{-\alpha x^2}$ where N and α are constants.

(b) If $f(s)$ is the Laplace transform of $F(t)$, then find the nth derivative $f^n(s)$.

17. Set up partial differential equation for a $3-D$ heat flow and obtain the expression for gradient of heat flow.

18. Discuss in detail the applications of tensors in dynamics of a particle.

19. State and prove Schur's lemmas for the irreducible representations of a group.

20. Solve the Laguerre's differential equation.

$xy''+(1-x)y'+\lambda y = 0$ Where λ is a constant and obtain the values of the first four Laguerre polynomials.

AFF–1021

Sub. Code
4MPH2C2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Physics

ELECTROMAGNETIC THEORY

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Find the electrostatic force of electron with proton in hydrogen atom.

2. At what condition Poisson's equation become a Laplace's equation.

3. "Magnetic forces do not work" - justify this statement.

4. If $\vec{A}$ is the vector potential, what does the curl of $\vec{A}$ represent?

5. Is there a displacement current in a good conductor why?

6. What are the advantages of Gauge transformation?

7. State and explain Snell's law.

8. Compare TE and TM waves.

9. Define retarded potential.

10. How will you use an electric dipole as a receiving antenna?

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) State and prove Gauss's law.

Or

(b) Describe the electrostatic boundary conditions.

12. (a) State and explain Lorentz force law.

Or

(b) Obtain Ampere's law in magnetized materials.

13. (a) Write down all Maxwell's equation in differential form and explain their physical significances.

Or

(b) What is called Poynting vector? Explain the significance of it.

14. (a) Explain the propagation of plane electromagnetic waves through a linear, isotropic, homogeneous and stationary media.

Or

(b) Prove that TEM waves cannot occur in a hollow wave guide.

15. (a) Write a brief note on "Scattering by free electrons (Thomson Scattering)".

Or

(b) Obtain expression for vector potential in the case of magnetic dipole radiation.

Part C

$(3 \times 10 = 30)$

Answer any **three** questions.

16. Obtain the multipole expansion of potential for an arbitrary charge distribution. Also explain the same.

17. Compare electrostatics with magnetostatics.

18. Show that under coulomb gauge, the electromagnetic scalar potential is exactly the electrostatic potential while vector potential is described completely in terms of transverse current.

19. Derive all the Fresnel's equation for an wave incident obliquely at an interface formed by two dielectric media.

20. (a) What is dispersion? Explain its types.

    (b) Briefly explain the dispersion in gases.

AFF–1022

Sub. Code
4MPH2C3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Physics

THERMODYNAMICS AND STATISTICAL PHYSICS

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. What is the principle of increase of entropy?

2. Distinguish between entropy and enthalpy.

3. State Boltzmann – H theorem.

4. Is Maxwell's velocity distribution law applicable to non-ideal gas.

5. Distinguish between microstate and macrostate of a system of particles.

6. What is meant by grand canonical ensemble?

7. What are the conditions for fermi-dirac statistics?

8. Why fermions don't show Bose – Einstein condensation type of phase transition?

9. What is the zero point energy of an electron gas?

10. Why the one dimensional Ising model never exhibits terro magnetism?

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Explain the first law of thermodynamics and derive the equation for the same.

Or

(b) Derive the clausius clayperon equation.

12. (a) Derive Boltzmann transport equation.

Or

(b) Set up and solve Langevin equation to obtain the mean square displacement of the particles undergoing Brownian motion.

13. (a) State and prove Liouville's theorem.

Or

(b) Discuss and prove the law of equipartition of energy in classical statistical mechanics.

14. (a) Compare Bose-Einstein statistics and Fermi-Dirac statistics.

Or

(b) Derive the equation of state of an ideal Fermi gas.

15. (a) Show that a Bose gas in two dimensions does not under go a B.E. condensation at any non-zero temperature work out the chemical potential and the specific heat of the gas.

Or

(b) Explain Lardan's theory of diamagnetism.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Explain the various thermo dynamic potentials of thermo dynamic functions.

17. Deduce Maxwell – Boltzmann law for the distribution of molecules in a gas. Evaluate the constants α and β.

18. Derive the relations between statistical quantities and thermo dynamical quantities in micro canonical ensemble.

19. Derive the Sackur – Tetrode equation for the absolute entropy of a monoatomic ideal gas.

20. Derive Planck's black body radiation law using of Bose – Einstein statistics.

AFF–1023

Sub. Code
4MPH2E1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Physics

Elective — NANO SCIENCE

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. "One nanometer is a magical point on the dimension scale", Why?
2. Give Moore's first and second laws.
3. List any four bottom up approaches for synthesis of nanopowders.
4. Classify nanomaterials and give examples for them.
5. Write down any two diameter – dependent properties of nanotubes.
6. What is self-assembly?
7. Give some important properties of light.
8. What is photons?
9. What do you meant by thermal MEMS?
10. Why nanotech weapons are superior over nuclear weapons?

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) How nanoscale dimensions affect the structural properties of the material?

Or

(b) What are the mile stones in the evolution of Nanotechnology?

12. (a) With a neat sketch, explain mechanical milling process for synthesis of nanoparticles. List the advantages and disadvantages also.

Or

(b) What are natural nano particles? Give applications of nanoparticles.

13. (a) Explain the purification method of carbon nanotubes.

Or

(b) Describe the function of proteins in molecular nanoelectronics.

14. (a) Write short note on "New low cost energy efficient windows".

Or

(b) What is photonic crystals? Give its properties and uses.

15. (a) Write short note on nanofabrication.

Or

(b) What are the types and applications of MEMS?

Part C (3 × 10 = 30)

Answer any **three** questions.

16. (a) What are the different types of Nanotechnology? Explain.

    (b) Explain the role of bottom up and top down approaches in Nanotechnology.

17. (a) What are the characteristics of nanoparticles that should posses by any fabrication technique?

    (b) Explain Sol-Gel synthesis for producing nano materials with a neat sketch.

18. (a) Describe in detail DNA analyser as Bio chip.

    (b) Explain the function of DNA and explain in detail about the DNA micro array.

19. Discuss the following :

    (a) Nanoholes

    (b) Imaging.

20. (a) What are quantum computers? Explain the experimental implementation of quantum computers.

    (b) Explain briefly the quantum information processing.

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AFF–1025

Sub. Code
4MPH2E3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Physics

Elective — COMMUNICATION ELECTRONICS

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. How can you obtain frequency modulation from phase modulation?

2. What is non-resonant antenna?

3. In what way is pulse code modulation different from other modulation systems?

4. When does the minimum shift keying occur?

5. What are the limitations of radar?

6. Give the principle of magnetron.

7. How does a step index fiber vary from a graded index fiber?

8. What is the purpose of cladding in the optic fiber?

9. What is meant by limits of visibility in relation to satellite communication?

10. What are GSM and CDMA?

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Explain the detection of AM signals in the presence of noise.

Or

(b) What is a half wave dipole? Draw its radiation pattern.

12. (a) Describe the functional blocks in a digital communication system.

Or

(b) What is PWM? How are PWM waves generated and demodulated?

13. (a) How to detect microwaves from IMPATT and TRAPATT diodes? Explain briefly?

Or

(b) Derive radar range equation.

14. (a) Mention the different types of couplers and their uses.

Or

(b) Discuss the parameters on which dispersion in an optical fiber depends. At which wave length zero dispersion in achieved.

15. (a) Write short note on MARISAT satellites.

Or

(b) Describe the account of up link and down link power across a satellite.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Analyze the frequency spectrum of a sinusoidal FM wave and hence derive the average power in sinusoidal FM wave.

17. Explain Frequency Shift Keying (FSK). How is FSK realized? Explain with wave forms.

18. Explain following types of radar

    (a) Pulse and CW radar.

    (b) MIT and auto tracking radar.

19. Explain the various types of splicings and describe the working of fusion splicing and its advantages.

20. Discuss the following in detail :

    (a) Analog cellular systems.

    (b) Digital cellular mobile systems.

AFF–1026

Sub. Code
4MPH3C1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Physics

QUANTUM MECHANICS

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. State limitations on the wave function.

2. Show that the momentum operator is Hermitian.

3. What is meant by zero point energy?

4. Define a Rigid Rotator.

5. What is Stark effect and Zeeman effect? Explain.

6. Write down the Hamiltonian for the ground state of the helium atom.

7. Prove : $[\vec{L}^2, L_x] = 0$ and $[L_+, L_-] = 2\hbar L_z$.

8. Define scattering cross section.

9. Write down the Pauli's spin matrices.

10. Define absorption, spontaneous emission and induced emission.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Give the physical significance of the wave function. What are the properties of the wave function?

Or

(b) Derive the equation of continuity in Quantum Mechanics. Interpret position probability density.

12. (a) Solve the Schroëdinger's wave equation for a particle in a one-dimensional box. Normalize the wave function.

Or

(b) Solve the solution for the radial equation of Hydrogen atom.

13. (a) Explain WKB approximation method, its principle and obtain an approximate solution for the wave function.

Or

(b) Describe Zeeman effect without electron spin and find an expression for the first order energy correction. Draw the transition diagram for the first three levels.

14. (a) Explain Stern-Gerlach experiment. What are its limitations?

Or

(b) Describe addition of angular momenta and obtain expression for Clebsch-Gordan coefficients.

15. (a) Derive an expression for the spin-orbit coupling energy for the Dirac's equation.

Or

(b) Starting from the relativistic linearised Hamiltonian in an electromagnetic field, explain the phenomenon of "Zitterbewegung" motion of the electron.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. (a) Derive the Schroedinger's time dependent wave equation. Deduce the time independent equation from it for stationary states. (6)

(b) Explain Dirac's Bra and Ket Vector notation. Describe dual space and change of basis vector. (4)

17. Solve the Schroedinger's wave equation for rectangular potential barrier. Find expression for reflection and transmission coefficient. Explain the Tunneling effect.

18. Explain Time dependent perturbation theory for the first order perturbation. Give physical interpretation. Obtain Fermi's Golden rule.

19. Use the Born approximation to get the differential scattering cross-section for

(a) spin less particle of energy E by a potential given by

$$V(r) = -V_o \quad \text{for } r > a$$
$$V(r) = 0 \quad \text{for } r < a$$

and (6)

(b) $V(r) = -g^2 \dfrac{e^{-\alpha r}}{(r)}$. (4)

20. (a) Deduce Klein-Gordon equation applicable exclusively for spinless particles and not for electrons. Can we use this equation for strong interacting particles such as π-Mesons and K-Mesons? (6)

(b) Obtain probability density and current density for the Dirac's equation for a free particle. Interpret the result. (4)

AFF–1027

Sub. Code
4MPH3C2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Physics

ATOMIC AND MOLECULAR PHYSICS

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. What is meant by hyperfine structure?

2. Explain the exchange symmetry of wave functions.

3. Define covalent and Ionic interactions.

4. Define absorption, spontaneous emission, induced emission.

5. Define symmetric top molecules, its two types with examples.

6. What are hot bands? Why are they called so?

7. What are overtone and combination vibrations?

8. What is predissociation?

9. What is chemical shift and coupling constant?

10. Write the applications of ESR.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Describe the main features of Alkali spectra and discuss the effect of spin orbit coupling on them.

Or

(b) Discuss $\vec{L}-\vec{S}$ and $\vec{j}-\vec{j}$ coupling schemes for two valence electron system.

12. (a) Describe stark effect and apply it to Hydrogen atom to find the splitting of energy levels and wave functions.

Or

(b) Explain population inversion and obtain Schawlow-Towne's condition for threshold oscillations in lasers.

13. (a) Deduce the rotational constant for a rigid diatomic molecule.

Or

(b) What are group frequencies in I.R. spectroscopy? How are they calculated for molecular bonding? What factors affect group frequencies?

14. (a) Explain the electronic spectra of diatomic molecules.

Or

(b) Explain in detail the nature and characteristics of Raman spectrum. Describe the Vibrational Raman spectra.

15. (a) Describe a ESR spectrometer with a block diagram, its working and applications.

Or

(b) What is Larmor precession? Derive an expression for frequency of precession. Write a note on spin flipping and Resonance.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. (a) For the two electron atom find an expression for the total energy applying schroëdingers wave equation system. (6)

(b) Describe Stern-Gerlach's experiment. (4)

17. Explain the quantum mechanical treatment of Normal Zeeman effect, Anomalous Zeeman effect and Paschen-Back effect.

18. (a) Obtain the energies of a diatomic vibrating rotator. Explain P and R branches. (6)

(b) The fundamental for HCl is centered at 2880 cm^{-1}. If the inter nuclear distance is 1.226 Å, calculate the wave number of the first two lines of each of the 'R' and 'P' branches of HCl. (4)

19. Discuss elaborately the rotational fine structure of electronic vibrational spectra.

20. Discuss the technique, principle and experimental methods of N.M.R. spectroscopy.

AFF–1028

Sub. Code
4MPH3C3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Physics

NUCLEAR AND PARTICLE PHYSICS

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. List out the characteristics of nuclear forces.
2. Give an account of nuclear stability.
3. Name the scintillator used for the detection of particles and photons.
4. What are the advantages of nuclear emulsion technique?
5. Define multiplication factor and state its values for fassion chain reaction.
6. Name the liquid coolants and gas coolants used in a reactor. What is pinch effect?
7. What are stripping and pick-up reactions?
8. Define threshold energy for nuclear reaction.
9. Explain elecromagnetic interaction.
10. Define parity, time reversal and charge conjugation symmetry operation.

Part B $(5 \times 5 = 25)$

Answer **all** questions by choosing either (a) or (b).

11. (a) Write a note on optical model.

Or

(b) Give an account of liquid drop model of the nucleus.

12. (a) Explain how Ionization chamber is used to detect nuclear radiations.

Or

(b) Write a note on internal conversion and nuclear isomerism.

13. (a) Describe cyclotron and find an expression for the radii of successive orbits.

Or

(b) Write a note on mass and energy distribution of fission fragments.

14. (a) State and explain the assumptions of the compound nucleus hypothesis.

Or

(b) Discuss in detail the partial wave analysis for nuclear reaction cross sections.

15. (a) Explain the law of conservation of Baryons.

Or

(b) Write a note on :

(i) Strange particles

(ii) Gell-mann Nishijjima scheme.

Part C

$(3 \times 10 = 30)$

Answer any **three** questions.

16. (a) Discuss the neutron-proton scattering at low energy. Find an expression for the total scattering cross-section for an S-wave.

    (b) Show how the assumption of spin-dependence of nuclear force can explain the experimental results.

17. Give a brief account of Fermi's theory of β decay.

18. (a) Explain briefly about nuclear fusion.

    (b) Describe about proton-proton chain reaction and carbon-nitrogen cycle. Derive the four factor formula for a thermal nuclear reactor.

19. Derive Breit and Wigner single level formula for scattering and absorption cross-sections in the vicinity of a resonance observed in neutron reactions.

20. Explain about :

    (a) Non-conservation of parity in weak interactions

    (b) C-P violation in neutral K-Meson decays.

AFF–1029

Sub. Code
4MPH3E1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Physics

***Elective* – ENERGY PHYSICS AND ENVIRONMENTAL SCIENCE**

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What are primary energy sources? Give examples.

2. What are the two main factors for the wind to blow?

3. Define altitude angle 'α' and Zenith angle 'θ_z'.

4. Name the main components of flat plate solar collector.

5. Name the various types of concentrating collectors.

6. List the advantages of flat plate collectors.

7. Name the main types of fuel cells.

8. Name the basic photovoltaic system for solar power generation.

9. What are the sources for air pollution?

10. Name the various classification of water pollution.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) List out the advantages and disadvantages of biomass as energy source.

Or

(b) Explain the basic principle, construction and working of a wind mill. Explain wind power in wind energy utilization.

12. (a) Explain Angstrom compensation pyrheliometer with a diagram.

Or

(b) Evaluate transmittance-absorptance product γ, α for a solar collector.

13. (a) Describe a point focus solar collector (paraboloid).

Or

(b) What are the disadvantages of concentrating collectors over flat-plate type solar collector?

14. (a) Find an expression for open circuit voltage in a solar cell.

Or

(b) List out the advantages and drawbacks of photo voltaic cells.

15. (a) List out the various methods of reducing air pollution.

Or

(b) What are the biological effects in human body as a result of water pollution?

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Explain briefly about :

    (a) Open cycle OTEC system and (5)

    (b) Closed cycle OTEC system with a schematic diagram. (5)

17. (a) Describe the construction and working of flat plate liquid heating collector, with a diagram. (5)

    (b) Explain the measurement of incident angle 'θ' and day length t_d. (5)

18. Explain briefly about :

    (a) Concentrating collector-non focusing type (5)

    (b) Compound parabolic concentrator. (5)

19. (a) State the advantages of fuel cells. (5)

    (b) Describe an Ion exchange membrance fuel cell and Molten carbonate fuel cell. (5)

20. Explain briefly with a diagram the following devices to reduce air pollution : $(5 \times 2 = 10)$

    (a) Electrostatic precipitators (wet and dry types)

    (b) Fabric filters

    (c) Wet scrubbers

    (d) Cyclones

    (e) Incinerators.

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AFF–1030

Sub. Code
4MPH3E2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Physics

Elective — MEDICAL PHYSICS

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define Inspiratory Lung Capacity (IC) and Functional Residual Lung Capacity.(FRC).

2. Sketch a typical ECG wave and explain the curves.

3. What is Doppler's effect?

4. Define phon loudness of sound.

5. State the different units for measuring light.

6. Give the medical uses of visible light.

7. What are the characteristics of X-Ray?

8. Define effective dose and equivalent dose.

9. State Bernoulli's principle.

10. State the functions of blood.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) What are the muscles that are used for inspiration and forced expiration? Explain briefly.

Or

(b) Write a brief note on electro oculogram.

12. (a) Discuss the physiological effects of ultrasound in therapy.

Or

(b) What are the different parts of hearing aids? Explain.

13. (a) Determine the refractive index of eye chamber water if the ray angle of draw is 45°.

Or

(b) Discuss the applications of microscopes in medicine.

14. (a) Describe the methods of production of X-ray beams.

Or

(b) Write notes on Air Kerma rate constant and reference Air Kerma rate.

15. (a) What are the major components of the cardiovascular system? Explain them briefly.

Or

(b) Explain about the different lead configurations used in ECG.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. (a) State the reasons for spherical aberration of an emmetropic eye and specify the methods of rectification.

    (b) How does the refraction of an emmetropic eye change while considering the prevailing colour constituent during the day?

17. (a) Describe the physical nature of outer ear function.

    (b) Outline the general properties of sound.

18. (a) What causes the unsharpness of X-ray image?

    (b) A 70 kg human body weight person was exposed to radiation for 25 minutes at a constant dose rate. The total amount of absorbed energy is 21 joules. Calculate the average dose to the body in mGY and the dose rate in mGY/s.

19. (a) Draw a simple scheme of blood circulation and mark in it the resistance and capacity systems as well as pressure conditions.

    (b) Systolic blood pressure is 20 kPa, while diastolic blood pressure is 14 kPa, the mean blood velocity in the aorta is 140 cm/s, the heart rate is 150 beats/min, the overall heart work in one systole is 32 joules. Calculate the cardic output in minute.

20. (a) Write a brief note on the occurrence of electric and magnetic signals from the heart.

    (b) At maximum in halation the pressure in the lungs is 1 kPa. Calculate the oxygen content partial pressure in lungs after maximum inhalation of air. (Air contains approximately 18% of oxygen).

AFF–1031

Sub. Code
4MPH3E3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Physics

Elective — MICROPROCESSORS AND MICROCONTROLLERS

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. What is the function of $\overline{IO}/\overline{M}$ signal in 8085?

2. List out the four instructions which could control the Interrupts of 8085 microprocessor.

3. Distinguish between memory mapped I/O scheme and peripheral mapped I/O scheme.

4. Name the major components of Keyboard/Display interface 8279.

5. Give the format of PSW register of 8051.

6. Compare microprocessor with microcontroller.

7. Explain the functions of the pin PSEN of 8051.

8. How the program memory is organized in 8051 based system?

9. Name the various registers used in 8051.

10. Define program counter and what is the significance of DPTR.

Part B (5 × 5 = 25)

Answer **all** questions choosing either (a) or (b).

11. (a) Explain PUSH and POP instructions in detail.

Or

(b) Describe the markable and non-markable interrupts available in 8085.

12. (a) Explain synchronous and asynchronous data transfer schemes.

Or

(b) Draw the block diagram of 8255 and explain its working. What is a control word?

13. (a) Name the special functions registers available in 8051.

Or

(b) Explain the interrupt structure, SFR and timers of 8051.

14. (a) Explain in detail about the programming of 8051 timer.

Or

(b) With a diagram explain how 8051 is interfaced to external memory.

15. (a) Describe the application of 8051 micro controller in pressure measurement.

Or

(b) Explain the usage of 8051 in D/A converter.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. (a) Explain the various addressing modes of 8085. (8)

(b) Write a brief note on status flag. (2)

17. (a) Draw the block diagram of programmable interrupt controller 8259 and explain its operations. (7)

(b) Write a note on Interrupt driven data transfer scheme. (3)

18. (a) What are the different addressing modes of 8051 microcontroller? Explain each with suitable examples. (7)

(b) Explain the structure of TCON, TMOD and PCON registers. (3)

19. (a) Explain the instruction set of 8051 microcontroller. (8)

(b) Write about the JUMP instructions present in 8051. (2)

20. (a) Explain in detail the seven segment display interfacing with 8051. (8)

(b) Explain about IP and IE registers. (2)

AFF–1032

Sub. Code
4MPH3E4

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Physics

Elective – MODERN OPTICS AND LASER PHYSICS

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Classify the electromagnetic spectrum as 'wave' region, 'optical' region, 'ray' region as per frequency, wavelength and quantum energy.

2. What is an evanescent wave?

3. Explain about division of wavefront and division of amplitude.

4. Define coherence time and coherence length.

5. State Green's theorem.

6. Explain recording amplitude and phase in interferometric recording.

7. What is stimulated Raman scattering.

8. Define an optical parametric amplifier.

9. A two level pumping scheme cannot be used in laser. Why? Explain.

10. Explain 'Spiking' in ruby laser. Write four applications of lasers in industry.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Explain circular and elliptical polarization.

Or

(b) Describe the method of observation of Doppler's effect with a laser.

12. (a) Explain the principle of linear superposition.

Or

(b) Describe Michelson's interferometer and discuss the theory of partial coherence and visibility of fringes.

13. (a) Derive the integral theorem of Helmholtz and Kirchhoff.

Or

(b) Explain the recording and reconstruction of images in a Gabor Holvgram.

14. (a) Explain in detail about the second harmonic generation.

Or

(b) Write a brief notes on :

(i) Kerr-induced self focusing

(ii) Plasma self focusing

(iii) Accumulated self focusing in hollow wave guides.

15. (a) Explain the construction, action, energy band diagram of a semi conductor diode laser.

Or

(b) Describe the construction, action and uses of He–Ne laser with the energy level diagram.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. (a) Discuss reflection and refraction of light at a plane boundary and obtain snell's law. (6)

(b) Explain amplitudes of reflected and refracted waves and derive Fresnel's equations. (4)

17. (a) Explain the theory of multilayer films and find expressions for reflection (r) and transmission (t) coefficient. Show that the sum of reflectance 'R' and transmittance 'T' is unity. (6)

(b) Write a note on anti-reflecting films. (4)

18. (a) Explain Leith-upatnieks hologram and about recording and reconstruction of images. (6)

(b) Write a note on high resolution volume imagery. (4)

19. Explain Franken's experiment. Derive Manley-Rowe relations.

20. (a) Derive 'Schawlow-Towne's condition' (threshold condition) for laser oscillation. (6)

(b) Write briefly the applications of laser in defence and communication? (4)